

Element I

Test and Evaluate The Prototype

Problem Statement

Reports indicate that American homeowners spend long hours each fall removing leaves from their yards, with 42,000 sustaining recurring neck, shoulder, and back injuries as a result per year. The significant time required to complete leaf removal not only causes physical strain but also reduces time available for other tasks.

The objective of this project is to design, construct, and evaluate a lawn-sweeper prototype that saves time and reduces physical strain while efficiently collecting and storing leaves.

Introduction

The purpose of testing the lawn sweeper prototype is to validate design predictions and evaluate performance against predetermined criteria. Testing addresses key features of the machine, including whether the system performs as intended, is reliable, and produces repeatable results.

A test is considered successful if:

- The results meet defined design specifications
- The data collected is accurate and consistent
- The testing procedure is repeatable

Build Process

The construction of the lawn sweeper prototype followed a structured process based on prior design development and research into mechanical leaf-collection systems.

Because of how big the prototype was designed to be. Only half of the product was printed and assembled.

Step 1: Component Fabrication

All custom components, including the body, hood, intake wheel, claws, handle, and frame, were designed in CAD software and fabricated via 3D printing with PLA.

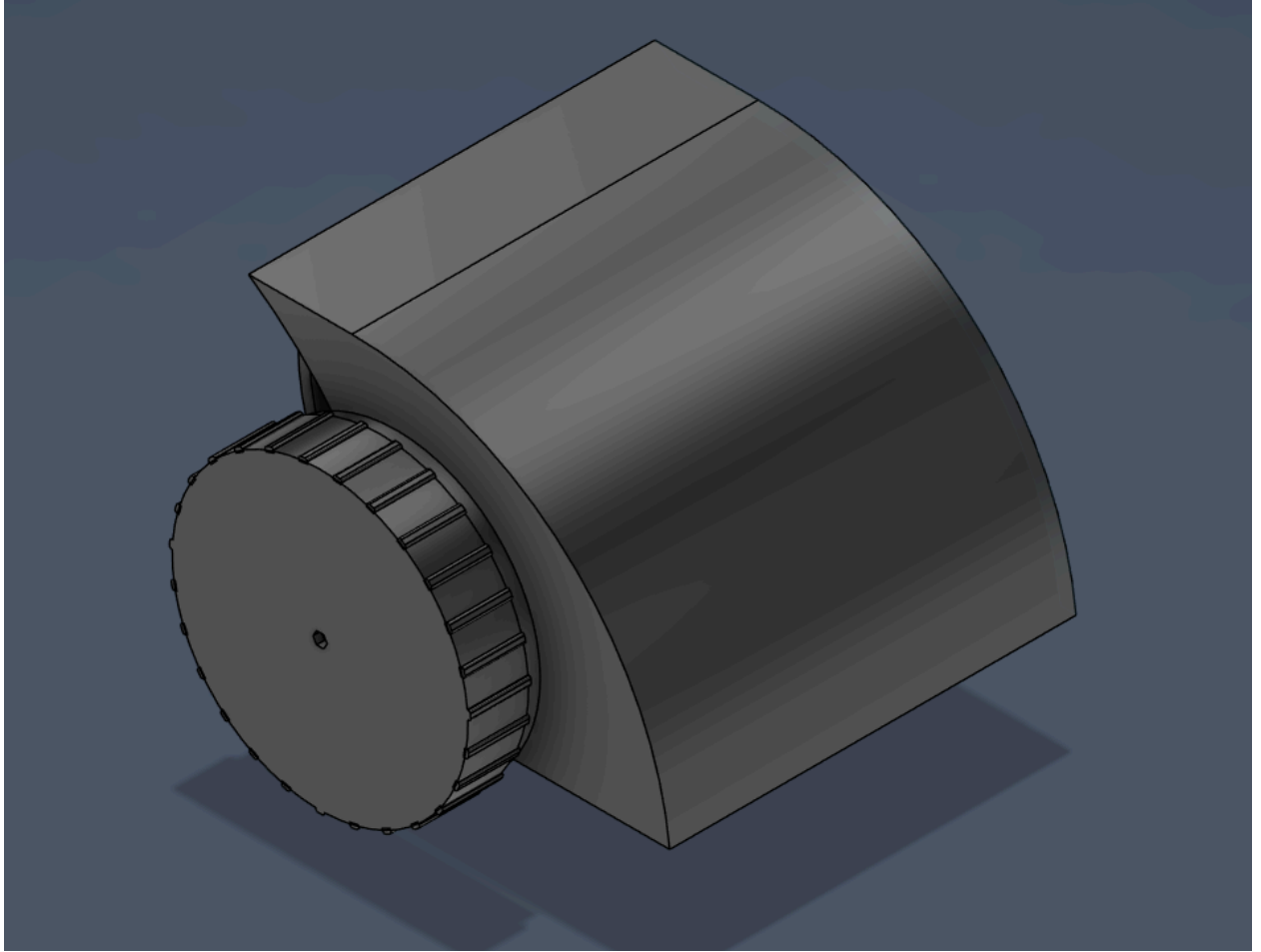


Figure I.1: CAD design of product hood and wheels

Following printing, components were inspected and sanded to ensure dimensional accuracy and proper fit.

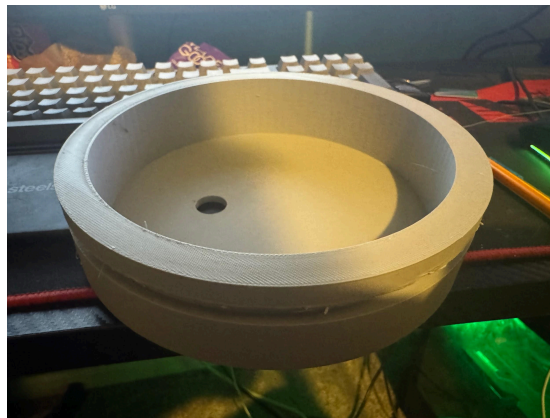


Figure I.2: 3D Printed Wheels

Step 2: Hood and Intake Assembly

The intake system was assembled by inserting bearings into designated mounting points on a screwed-in bearing holder. These bearings support the axle and enable smooth rotational motion.

The intake wheel and claws were attached to the axle, forming a rotating mechanism that lifted and guided leaves into the storage system.

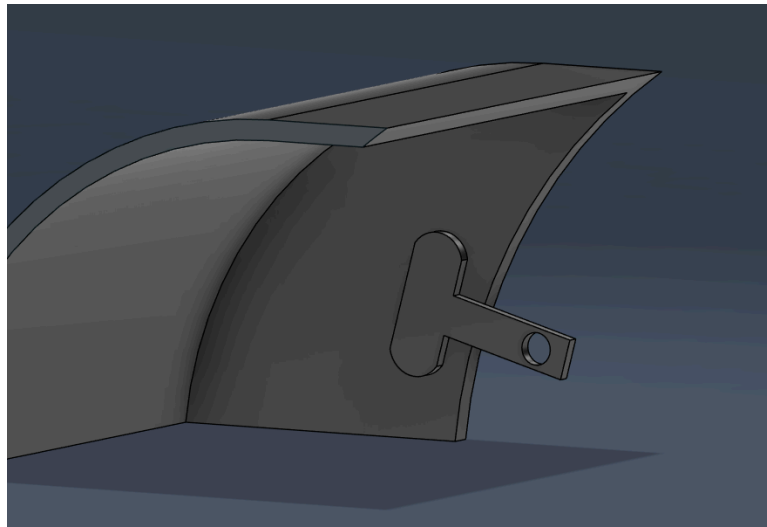


Figure I.3: Hood 3D Design Rotating Mechanism

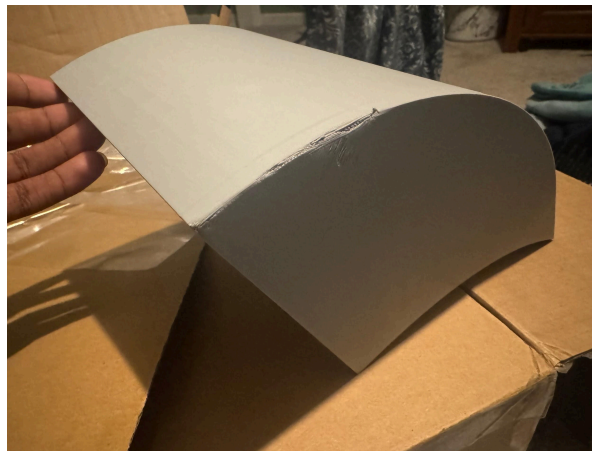


Figure I.4: 3D Printed Hood

Step 3: Body and Wheel Assembly

The axle was inserted through the central body structure and connected to two wheels. Shaft collars were used to secure components and prevent unwanted movement.

Wheel size and placement were selected to improve traction, stability, and ease of movement across uneven terrain.



Figure I.5: Assembled Intake System

Step 4: Storage System Installation

A frame was attached to the rear of the device to support a leaf collection bag. This system enables continuous collection and storage during operation.

The design allows for easy removal and disposal of collected leaves.

Step 5: Final Assembly

All subsystems were integrated into the final prototype. The handle was attached to provide user control, and all structural connections were secured.

The completed prototype operates through manual input, with mechanical motion transferred to the intake system.

Incremental Testing

Incremental Test #1: Wheel and Axle Rotation

This test evaluated whether the wheels and axle rotated smoothly without excessive friction and sufficiently to move the machine forward.



Figure I.6: Assembled Wheel with gear components

Result: Successful. Rotation was consistent and unobstructed.

Incremental Test #2: Gear System

This test verified that rotational motion from the wheels was effectively transferred to the intake mechanism.

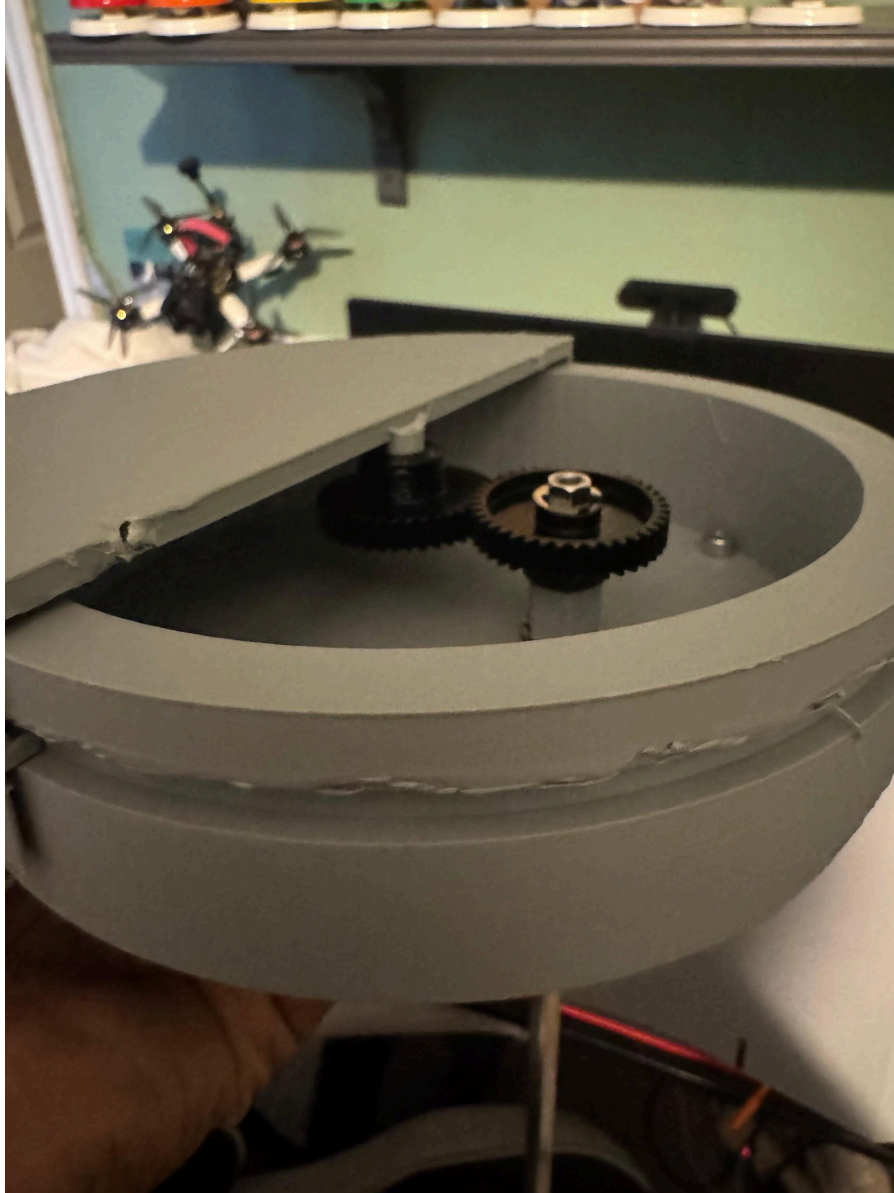


Figure I.7: 3D Printed wheel with gear components

Result: Successful. The intake system rotated as expected, in sync with the wheels.

Incremental Test #3: Intake Mechanism

This test assessed the intake claws' ability to collect leaves.

Result: Partially successful. Some leaves could be picked up, but others were displaced rather than lifted, indicating inefficiencies in the intake design.

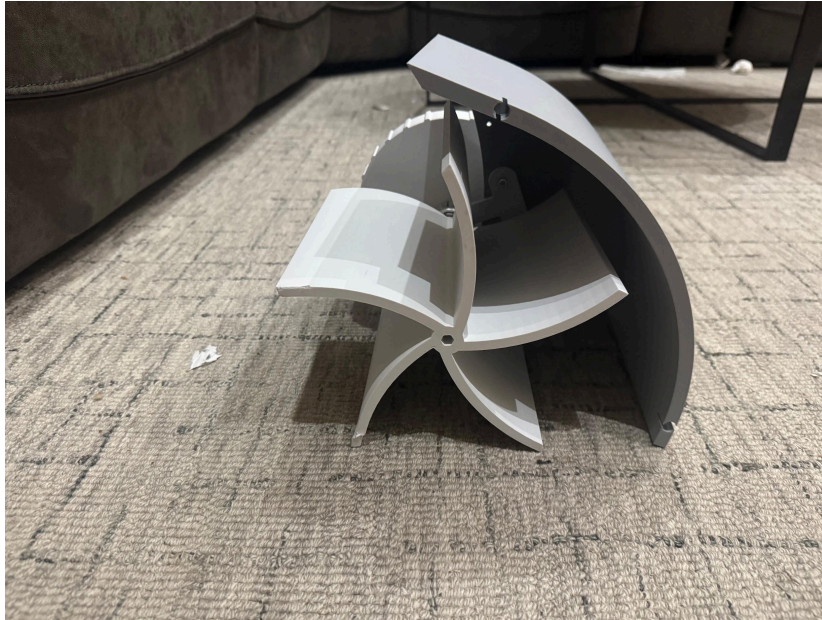


Figure I.8: Fully assembled intake system with wheels

Test Process

Following construction, the prototype was tested on these criteria:

1. Efficiency
2. Ergonomics
3. Performance in Natural Conditions

Each test was conducted multiple times to ensure reliability and consistency of results.

Test #1: Efficiency

Test Procedure

1. Distribute leaves evenly across the testing area

2. Start the timer and operate the machine for a set period.
3. Measure the total volume of leaves at the end of the test
4. Calculate Efficiency in gallons/hour.
5. Record all data on the accompanying datasheet.

Results

Test #	Mass Leaves(lb)	Volume Collected (gal): .03 lb/gal	Time (min)	Efficiency (gal/hr)
1	1.45	4.83	10	29.0
2	1.6	5.33	10	32.0
3	1.35	4.50	10	27.0
4	1.50	5.00	10	28.0

Table I.9: Efficiency Test Results

Reflection

This test generally yielded consistent results across trials, indicating that the test procedure was designed to be repeatable. It was necessary to approximate the density of the leaves to calculate the volume using mass, as measuring volume directly is difficult.

Towards the later trials, the prototype began to experience mechanical issues, and it required many adjustments between tests. This suggests that the product may not be as durable or stable as expected because of the PLA material.

Recommendations

The intake claws could be redesigned or repositioned to improve leaf-collection efficiency and prevent leaves from being pushed forward instead of lifted. Using more durable materials or reducing the thickness of the major 3D-printed parts may also help lighten the prototype.

Test #2: Ergonomics

Test Procedure

1. All participants are familiar with how to use the prototype.
2. Test the product, letting participants use the prototype in the designated lawn area.
3. Have all participants fill out the survey
4. Repeat the test with at least 5 participants
5. Record all data and survey responses on a spreadsheet

Results

Participant	Comfort Rating (1–5)	Feedback
1	2	Handles felt loose, sometimes falling out while using the product. It was annoying having to bend down and put it back in over and over. The handlebars were also too low, so it would eventually cause a sore back from extended use.
2	3	It was rough when pushing forward. The axle components are getting stuck between the ground and are not able to rotate properly.
3	4	Some leaves didn't always get swept up, leaving behind a trail of scattered leaves behind as the product was being pushed. Overall, the product was easy and self-explanatory to use.
4	3	Sometimes the product would get stuck and not move, requiring a lot more force to move the machine. (This happened because the axle components were pressing against the ground)
5	3	Wheels weren't moving at first because of how close they were to the hood of the product, but overall, pretty easy to use despite the short handlebar height
6	4	While using the product, one of the handles fell out once but the product still worked pretty well and picked up most leaves.

Table I.10: Ergonomics Test Results

Reflection

Throughout the test, the product kept having issues with the same components mentioned above. Even after redesigns and modifications, the same issue kept occurring. 3D printing with the PLA material wasn't as strong as it needed to be for the users, and having to re-print components cost more time.

Participant number 6 had the product when the most modifications had been made, but still faced issues with the wheels due to sizing and the material. Another material for 3D printing should be selected if the lawn sweeper moves further into production.

Another issue with the test was the inconsistencies among the test subjects. Participants varied in attributes like height and physical abilities, giving it inconsistent results despite being the same product. For example, taller participants may have had a lower comfort rating than the participants who may have had the height that is perfect for comfort with the product. This makes the results of this test inconsistent but also more representative of the real world.

Recommendations

For the future, adjustments can be made to the product in terms of making the handlebars adjustable so that they can be adjusted to a comfortable position for all users. Additionally, adding more spacing between the components of the product would also be beneficial in order to minimize the amount of jamming in the product from other components.

Test #3: Natural Conditions

Test Procedure

The prototype was tested under varying environmental conditions, including dry and wet leaves and debris. Using paper leaves to recreate the environment.

Results

Condition	Total Leaves	Leaves Collected	% Collected	Notes
Dry	50	28	56	Was able to pick up dry leaves with moderate consistency, but still needed to dig deeper into the ground to pick up everything.

Wet	50	16	32	Cannot pick up wet leaves very well. They mostly just stick to the intake or slide off and stay on the ground. So it should dig into the ground more
Debris	50	38	76	Crumpled up the paper to simulate bigger debris, and it was able to capture them with ease.

Table I.11: Natural Conditions Test Results

Reflection

The results from Test #3 show that the lawn sweeper performs inconsistently across different types of material collected. This product was most effective when handling larger debris. This suggests that the current intake mechanism is well-suited to bulkier items; however, it is slower to pick them up than leaves.

Performance dropped when picking up dry leaves, and especially wet leaves. For dry leaves, there was moderate success, but they struggled to collect those firmly pressed into the ground. This indicates that the current design does not reach low enough to collect all items.

Recommendations

- **Increase intake depth:**

Adjust the sweeper so it can dig slightly deeper into the ground. This would help collect more dry and wet leaves that may be embedded or stuck to the ground.

- **Add downward force**

Allowing the user to apply more pressure downward or adjust the height of the product could improve the amount of contact the intake has with the ground and improve collection effectiveness.

Evaluation of Test Results

The testing results show issues in the following parts:

- Materials: PLA components lack durability under a lot of force
- Build Quality: Minor misalignments reduce efficiency
- Design: Intake mechanism requires more efficient shaping and needs to be better sized

Conclusion

The lawn sweeper prototype demonstrates potential as a solution for efficient leaf collection while saving time and reducing physical strain. However, improvements in intake performance, durability, and ergonomics are required for full optimization.

Engineering design is an iterative process because the initial solutions rarely meet all requirements without being refined. Testing and evaluation during the build is very important for design improvements. For example, the lawn sweeper required many revisions throughout the building process, often due to the 3D printed parts printing incorrectly or not fitting as expected. Many parts had to be re-caded and reprinted because they would not fit properly or would be too weak to be used in the prototype due to the PLA material. The main issue for this came in the intake system. Originally, the design had a bunch of intake blades that would sweep in the leaves. However, the blades turned out to be very weak and broke a lot during testing, requiring them to be redesigned into plates so that they could be more durable and perform better.

The tests performed on the product gave valuable insight into how to move forward with the product for it to be mass-produced. For example, the ergonomics test gave valuable insight into the design of the handlebars, and how it would be better for the design to have an adjustable handlebar for comfort for users of varying heights. The material used to 3D print the design also needs to be changed to a stronger material to prevent it from breaking easily. By using a stronger material, the product's durability and lifespan will increase, making it more appealing to buyers.